SIM-02A supplement: sensitivity of guaranteed bands to the ambiguity criterion

Generated by `tools/sim02a_criterion_margin.py` from the resolved configuration in `results/sim02a/20261107T214846Z_c56c0048_31da01b/record.json` (config sha256 `c56c00480adbebce...`). Code commit `48c4cfc` (clean tree). The run directory is read, never written.

Software: picket_sim 0.1.0, python 3.13.18, numpy 2.5.4; run of record: picket_sim 0.1.0, python 3.13.18, numpy 2.5.4.

Self-check: the guaranteed and median bands recomputed at -3, -6 and -10 dB reproduce the run-of-record `thresholds_summary.csv` (156 values, relative tolerance 1e-09).

Narrowband DAS, error-free, horizon steering set (24 azimuths at 15 deg), observable-space lobes. Guaranteed band: latest f_SL to earliest f_amb(L) over the steering set (SIM-01 Rev A definition). These are criteria sensitivities, not requirements (D03).

1. Guaranteed band ratio at each criterion and at the criterion +/- 0.1 dB

ID	-2.9	**-3.0**	-3.1	-5.9	**-6.0**	-6.1	-9.9	**-10.0**	-10.1
A1	1.78	**1.77**	1.77	1.69	**1.69**	1.69	1.63	**1.62**	1.62
A2	1.78	**1.77**	1.77	1.69	**1.69**	1.69	1.63	**1.62**	1.62
A3	1.29	**1.28**	1.27	1.14	**1.14**	1.13	1.03	**1.03**	1.03
A4	1.07	**1.06**	1.06	empty	**empty**	empty	empty	**empty**	empty
SQ4-C	1.29	**1.28**	1.27	1.14	**1.14**	1.13	1.03	**1.03**	1.03
P6-C	2.54	**2.53**	2.53	2.38	**2.37**	1.65	1.28	**1.28**	1.27
P8-C	3.79	**3.78**	3.77	2.81	**2.80**	2.79	1.33	**1.32**	1.32
SQ4-E	1.29	**1.28**	1.27	1.14	**1.14**	1.13	1.03	**1.03**	1.03
P6-E	2.54	**2.53**	2.53	2.38	**2.37**	1.65	1.28	**1.28**	1.27
P8-E	3.79	**3.78**	3.77	2.81	**2.80**	2.79	1.33	**1.32**	1.32
V-C	1.07	**1.06**	1.06	empty	**empty**	empty	empty	**empty**	empty
V-D	1.07	**1.06**	1.06	empty	**empty**	empty	empty	**empty**	empty
V-DS	1.16	**1.14**	1.13	empty	**empty**	empty	empty	**empty**	empty

A large change across a 0.2 dB interval means the nominal band edge is set by a sidelobe plateau lying within 0.1 dB of the criterion, not by a lobe that rises steadily through it.

Knife-edge cases (guaranteed ratio changes by more than 10% of its nominal value across the +/- 0.1 dB interval): P6-C at -6 dB (2.38 at -5.9 dB, 1.65 at -6.1 dB); P6-E at -6 dB (2.38 at -5.9 dB, 1.65 at -6.1 dB).

2. Worst-steering peak sidelobe level across the guaranteed -3 dB band

Peak sidelobe level (PSL) relative to the mainlobe, maximum over the horizon steering set, at each grid frequency (30 points per decade) inside the guaranteed -3 dB band. Full values in the `_psl.csv` file.

Plateau: the worst-steering PSL where it is constant to 0.001 dB over at least 3 consecutive grid frequencies (longest such run), with its frequency span.

ID	-3 dB band (Hz)	Grid points	Lowest worst-steering PSL (dB)	Plateau (dB)	Plateau span (Hz)	Highest (dB)
A1	2,501 to 4,432	8	-11.30	-11.30	2,518 to 3,991	-4.64
A2	1,251 to 2,216	8	-11.30	-11.30	1,262 to 2,000	-4.49
A3	1,465 to 1,877	4	-11.23	none	n/a	-3.27
SQ4-C	1,382 to 1,769	3	-8.40	none	n/a	-3.66
P6-C	1,283 to 3,253	12	-20.76	-6.02	2,332 to 2,936	-4.01
P8-C	1,276 to 4,826	17	-21.73	-7.77	2,160 to 3,170	-4.11
SQ4-E	691 to 885	3	-8.30	none	n/a	-3.61
P6-E	642 to 1,626	12	-20.48	-6.02	1,169 to 1,471	-3.91
P8-E	638 to 2,413	17	-21.42	-7.77	1,082 to 1,589	-4.11
V-C	1,301 to 1,384	1	-3.20	none	n/a	-3.20
V-D	146 to 156	1	-3.75	none	n/a	-3.75
V-DS	159 to 182	1	-3.52	none	n/a	-3.52

Computed in 271 s.